

Hitarth Narvala

DEPUTY VICE PRESIDENT | DATA SCIENCE | PhD in NLP/IR

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Profile

Data science leader and NLP/Information Retrieval researcher with a PhD in Computing Science from the University of Glasgow and 14+ years of experience spanning data science and data engineering across financial services and healthcare domain. Currently Deputy Vice President (Data Science) at Kotak Mahindra Bank, leading deep learning and generative AI initiatives, translating research into business impact through hands-on model development and team leadership. Research expertise includes neural recommendation systems, knowledge-graph representations, information threading, information extraction and text classification.

Professional Experience

Deputy Vice President - Data Science

Kotak Mahindra Bank, India | Jun 2024 - Present

- Created Kotak's first deep neural network for cross-sell recommendation, replacing 40+ propensity models and delivering a 12–18% improvement in real-world conversions.
 - Hybrid recommendation framework that models customer product holdings, transactions and bureau data over time to recommend the most relevant banking products.
- Designed a generative AI (GenAI) explainability framework using a multi-agent architecture powered by large language models (LLMs) to interpret customer portfolios, explain model recommendations and guide relationship managers in customer communication.
- Currently designing a foundation model for Kotak Mahindra Bank.
- Managing direct-report data scientists in Senior to Lead Data Scientist role.

Research Database Manager - Full-stack Development

University of Glasgow, UK | Jan 2019 - Dec 2023

- Designed and developed web-based data collection tools and managed databases for research studies in the Population Health Research Facility.
- Added data analysis, online statistical visualisations and reporting to research data collection tools.
- Worked with researchers and business users to translate data requirements into effective database models.

Consultant - ETL Data Engineering

Exusia India Pvt Ltd, client: United Health Group, India | May 2017 - Aug 2018

- Led the ETL data engineering team and contributed technical expertise to data warehousing projects.
- Achieved 70–90% application runtime performance improvements through data-flow analysis and parallel processing.
- Applied Pharmacy Benefit Management expertise to requirements and design; mentored junior team members.

Technical Team Lead - ETL Data Engineering

Wipro Technologies, client: Citi Bank, Singapore and India | May 2012 - May 2017

Technical Team Lead (2015 - 2017); Senior Project Engineer (2014 - 2015); Project Engineer (2012 - 2014).

- Progressed from ETL development to technical team leadership, delivering data warehousing projects and resolving critical production issues across financial ledgers and electronic cards.
- Led Level 3 investigations, mentored developers, and worked with business users on requirements, project planning and training across Singapore, China, the USA and Mexico.

Education and Research

PhD in Computer Science

University of Glasgow, Glasgow IR Group, UK | 2019 - 2024

Research focus: Natural language processing (NLP), information retrieval (IR) and sensitivity review

Supervisors: Prof. Iadh Ounis and Dr. Graham McDonald

- Developed IR and NLP methods to identify latent relations between documents and support human sensitivity review of personal and confidential information.
- Proposed novel methods for knowledge graph embeddings, document recommendation, information threading and text classification.
- The research is featured in various publications across top-tier international conferences, namely ACM SIGIR, EMNLP, ACM CIKM, ECIR and ACM CHIIR, along with a long article in the IP&M journal
- Assisted supervision of undergraduate and master's projects on LLMs, news recommendation, event extraction and sensitivity classification.
- Served as Local Chair and Program Committee member for ECIR 2024; organised the IR seminar series (2022–2023) in the School of Computing Science, featuring experts from the IR and NLP community.

MSc in Data Science, with Distinction | University of Glasgow, UK | 2018 - 2019

BTech in Computer Science, with Distinction | Amity University Rajasthan, India | 2008 – 2012

Awards

Kotak Mahindra Bank: Kotak Star of the Month (Aug 2025); The Disrupters - Team Award (Q4 FY 2026).

Winner, J.P. Morgan Code for Good hackathon (2018); Best Catalyst, Exusia–UHG (2017); Instant Karma, Wipro–Citi (2014); Bravo Choice, Wipro–Citi (2013).

Technical Skills

Data science: Machine learning, deep learning, NLP, information retrieval, recommendation systems, information extraction, generative AI, LLMs, multi-agent systems, model explainability, data visualisation.

Languages and frameworks: Python, PyTorch, TensorFlow, NumPy, pandas, scikit-learn, spaCy, Matplotlib, Java, SQL, Django, PHP, JavaScript, HTML, CSS.

Data and platforms: Ab Initio, data warehousing, ETL, Linux/Unix, shell scripting.

Certifications

NVIDIA: Fundamentals of Deep Learning for Computer Vision (2019); Fundamentals of Accelerated Computing with CUDA (2019). Oracle: SQL Certified Expert (2010).